

RMO(INDIA) – 2008

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let $\triangle ABC$ be an acute angled triangle; let D, F be midpoints of BC and AB respectively. Let the perpendicular from F to AC and the perpendicular from B to AC meet in N . Prove that ND is equal to the circum radius of ABC . [15]
 2. Prove that there exist two infinite sequences $\langle a_n \rangle_{n \geq 1}$ and $\langle b_n \rangle_{n \geq 1}$ of positive integers such that the following conditions hold simultaneously: [19]
 - (i) $1 < a_1 < a_2 < \dots$
 - (ii) $a_n < b_n < a_n^2$ for all $n \geq 1$.
 - (iii) $a_n - 1$ divides $b_n - 1$ for all $n \geq 1$.
 - (iv) $a_n^2 - 1$ divides $b_n^2 - 1$ for all $n \geq 1$.
 3. Suppose 'a' and 'b' are real numbers such that the roots of the cubic equation $ax^3 - x^2 + bx - 1 = 0$ are all positive real numbers. Prove that: [19]
 - (i) $0 < 3ab \leq 1$.
 - (ii) $b \geq \sqrt{3}$.
 4. Find the number of all 6 digit numbers such that sum of their digits is 10 and each of the digits 0, 1, 2, 3 occurs at least once in them. [14]
 5. Three non zero real numbers a, b, c are said to be in harmonic progression if $\frac{1}{a} + \frac{1}{c} = \frac{2}{b}$. Find all three terms of harmonic progression a, b, c of strictly increasing positive integers in which $a = 20$ and b divides c . [17]
 6. Find the number of all integer-sided isosceles obtuse angled triangles with perimeter 2008. [19]

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